Paleomagnetism of the Western Channel Diabase and Associated Rocks, Northwest Territories

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The stable remanent magnetization of the Western Channel diabase (1325–1785 m.y.) determined at 35 sites has a mean direction of 356, −50 ($\alpha_{95} = 6°$) and a corresponding pole at 9° N, 115° W ($A_{95} = 6°$). Studies of rocks bordering the diabase show that this magnetization was acquired at the time of initial cooling. Preliminary results from older sediments, diabase, and porphyry (1770 m.y.) give directions of opposite sign, with corresponding poles in the same general region. These results, and others recently published, allow the path of apparent polar wandering, relative to the Canadian Shield, to be constructed in a preliminary fashion for the middle and earlier Proterozoic. This path, together with that previously obtained by other workers for the late Precambrian, suggests that during the Proterozoic the pole moved through two cycles of a roughly sinusoidal path with a peak-to-peak amplitude of about 90° of arc and a period of several hundred million years.

La magnétisation rémanente stable du diabase de "Western Channel" (1325–1785 m.y.), déterminée à 35 endroits, a une direction moyenne de 356, −50 ($\alpha_{95} = 6°$) et un pôle correspondant à 9° N, 115° W ($\alpha_{95} = 6°$). Des études faites sur les roches entourant le diabase montrent que cette magnétisation a commencé au début du refroidissement. Des résultats préliminaires à partir des sédiments plus vieux, du diabase et du porphyre (1770 m.y.) donnent des directions de signe contraire, avec un pôle correspondant situé dans la même région générale. Ces résultats, et d'autres publiés récemment, permettent de construire une forme préliminaire de la trajectoire d'un pôle apparent divaguant, relativement au Bouclier Canadien, au Protérozoïque Inférieur et Moyen. Cette trajectoire, ainsi que celles observées précédemment par d'autres auteurs au Précambrian Supérieur, suggèrent que, durant le Protérozoïque, le pôle s'est déplacé de deux cycles grossièrement sinusoïdaux avec une amplitude de sommet à sommet de 90° d'arc environ et une période de plusieurs millions d'années.

Introduction

The sampling area is near Hornby Bay, an arm of Great Bear Lake (Figs. 1, 2, and 3). The geological sequence consists of the Cameron Bay and Echo Bay Groups (folded Proterozoic volcanics and sediments) intruded by granite, and unconformably overlain by flat-lying to gently dipping sandstones and dolostones of the Hornby Bay Group. All of these lithologies are intruded by diabase. On the basis of earlier descriptions, the volcanic and red bed units seemed to hold promise for paleomagnetic studies. However, red beds are rare, and the volcanics commonly massive, so that original attitudes can rarely be determined with confidence; consequently, only small collections of these rocks could be made. By contrast, the diabase was most suitable for sampling, and work was therefore concentrated on this lithology, and on host rocks adjacent to diabase contacts (to determine the time at which magnetization was acquired).

Field work was carried out in 1968 from a camp of the Geological Survey of Canada organized by A. Donaldson. Collections were made by E. Irving and A. Donaldson by boat. We are grateful to Andrew Miall, Alan Kohut, and Chin Yoon Chork for their assistance in

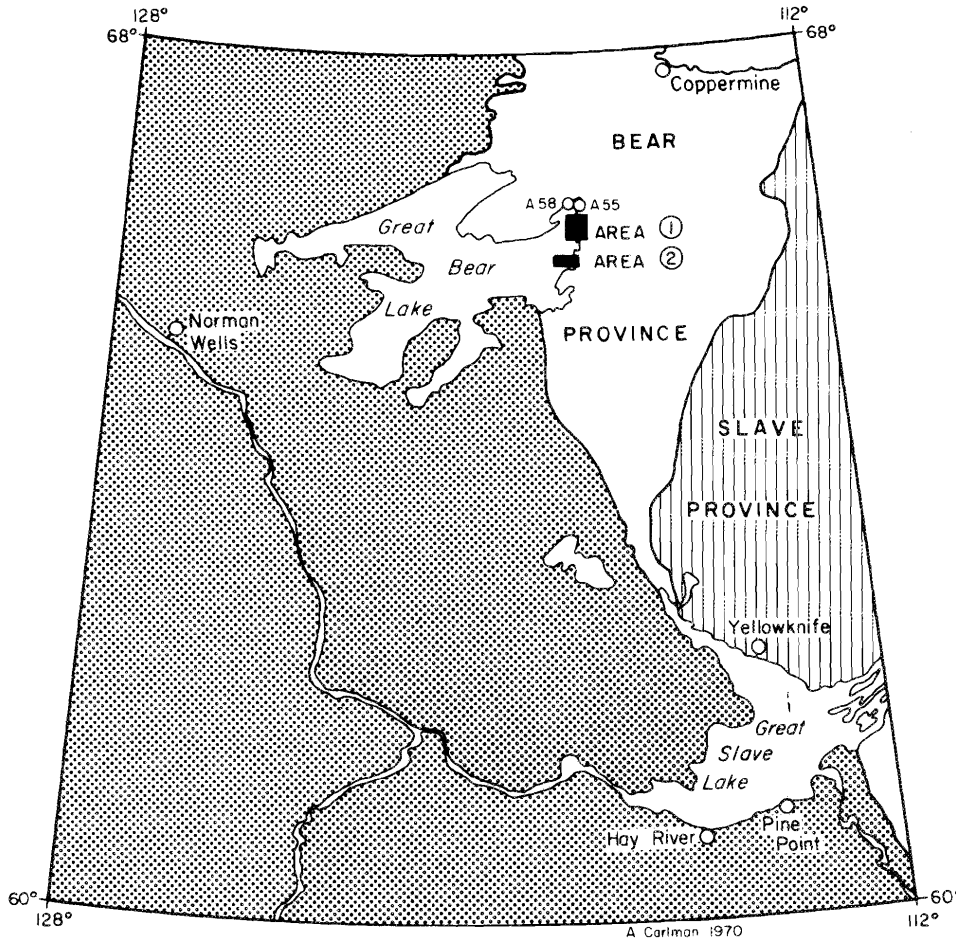


FIG. 1. Index map showing the location of the main sampling areas (see Figs. 2 and 3 for details) together with the location of two sites (58 and 55) outside these main areas. The Bear and Slave provinces are Precambrian. Phanerozoic rocks underlie the stippled area.

the field. Sampling was begun using a gasoline-powered drill (5 or 6 cores per site), but because of drilling difficulties, collections were completed by the hand-sample method (usually 3 samples per site). Where possible, two specimens were cut from each core or hand-sample, and the average of these pairs has been used as the basic statistical unit.

The magnetic measurements were made by J. Park and E. Irving. Measurements of remanence were made using a commercial spinner magnetometer, and demagnetization was carried was carried out using the apparatus described by Roy *et al.* (1971). Coercivities (median destructive field (m.d.f.) of anhysteretic remanent magnetization (a.r.m.) and saturation isothermal remanent magnetization (i.r.m.)) were measured using the procedures described by Park and Irving (1970).

Western Channel Diabase

Diabase occurs as: (1) sheet-like bodies on the peninsula between Hornby Bay and Western Channel (Fig. 2); (2) as easterly and north-northeasterly trending dikes that extend from Western Channel as far south as Port Radium (Fig. 3); and (3) as sheet-like bodies having arcuate outcrop (Port Radium sills, Fig. 3). Occurrences (1) and (2) have essentially the same direction of magnetization and

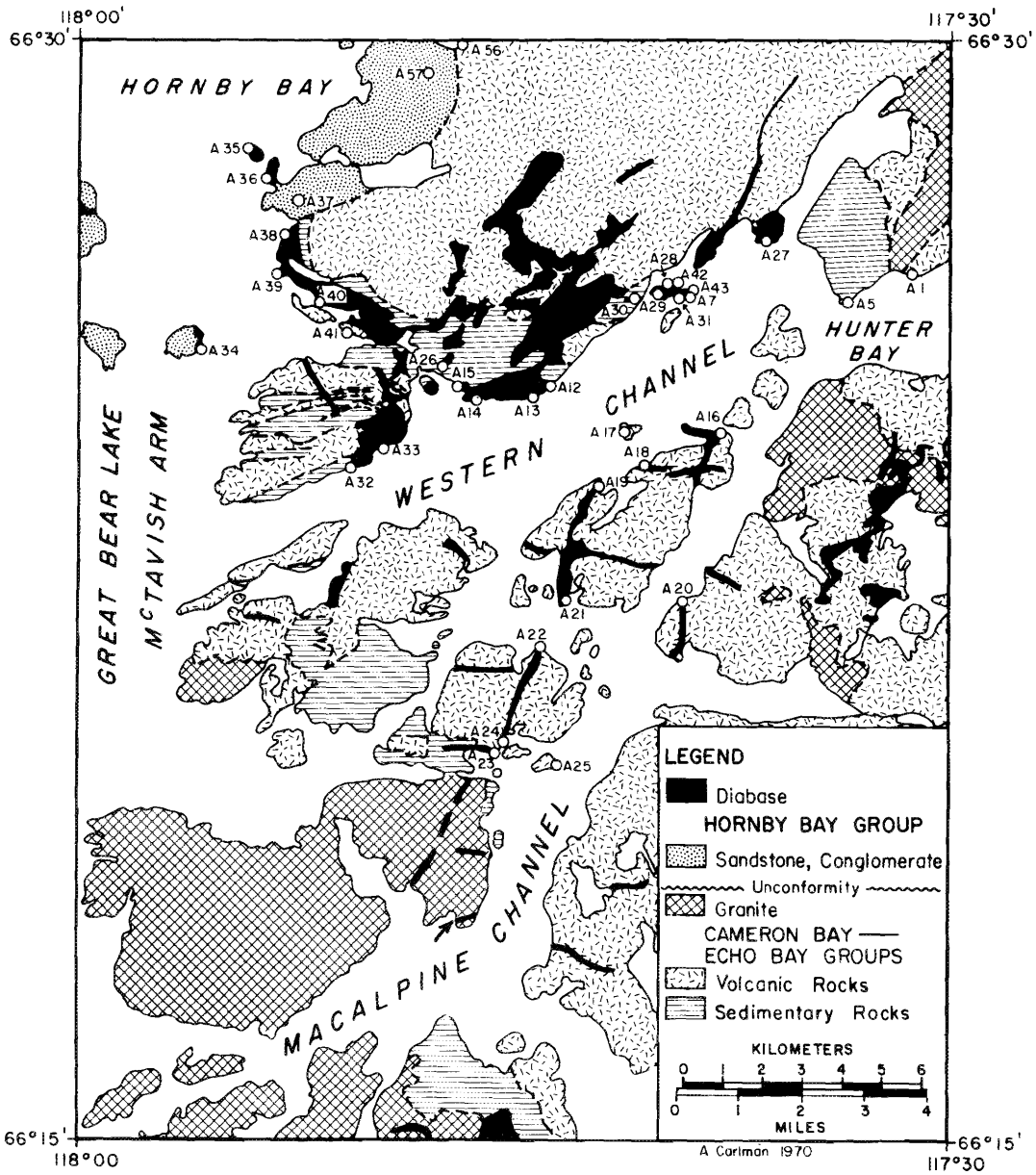


FIG. 2. Sampling area 1. Geology generalized after Feniak (1948). Sampling site numbers are prefaced by the letter A. This is not repeated in the text and tables for the sake of brevity. The site numbers have no stratigraphic significance. Arrow indicates the location of dated diabase sample (see text).

are referred to here as the *Western Channel diabase*. A biotite–hornblende extract from diabase (GSC 67–80 collected by W. F. Fahrig; see Fig. 2) gave an age of 1400 ± 75 m.y. by the K–Ar method (Wanless *et al.* 1970); because this sample is somewhat altered, this is regarded as a minimum age. The mean K–Ar age of granitic rocks in the vicinity (including those cut by Western Channel diabase) is 1785 m.y. (Stockwell 1970). The present estimate of the age of the diabase therefore lies in the range 1325–1785 m.y. In hand specimen the

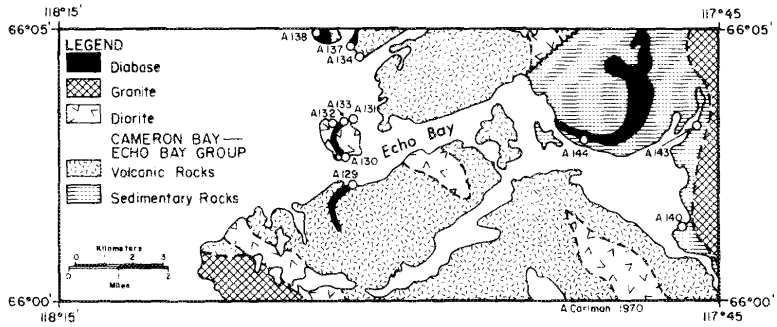


FIG. 3. Sampling area 2. Geology generalized after Kidd (1963). Port Radium is 1 km north of station A137.

diabase is very fresh with bluish gray to dark greenish gray color and a clean brittle fracture.

The diabase sheets were sampled at 26 sites (Fig. 2). All, except site 17 which was random, gave excellent data (Table 1). The directions initially are widely scattered, the average standard deviation exceeding 40° (Table 2), and the mean site directions have an almost random distribution (Fig. 4a). Typical alternating field (a.f.) demagnetization curves are shown in Fig. 5. Stable end-points were determined for each site and all samples were magnetically cleaned. Cleaning decreases the scatter, often in a dramatic fashion (Fig. 6), the average standard deviation being reduced to about 10° (Table 2). The mean site directions after magnetic cleaning are well grouped, and diverge by more than 100° from the direction of the present geomagnetic field (Fig. 4b, Table 1). Diabase dikes were sampled at 13 sites, 10 of which gave good data. Eight sites are located in what may be a single body extending from Western Channel to Port Radium (site 19 of Fig. 2 south to site 129 of Fig. 3). Two sites (16 and 20) are in a second dike of

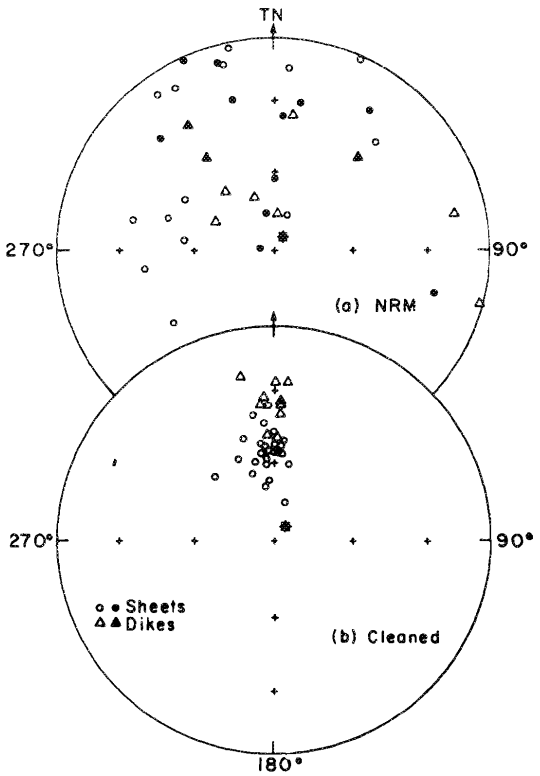


FIG. 4. Directions of remanent magnetization in the Western Channel diabase. Mean site directions are plotted before (a) and after (b) magnetic cleaning. Results from sheets and dikes are distinguished. Solid (open) symbols indicate positive (negative) inclinations. The primitive is the horizontal plane. The present earth's field is marked by a star.

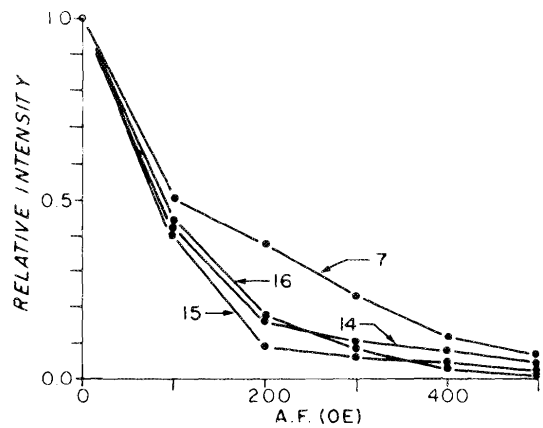


FIG. 5. Typical a.f. demagnetization curves for the Western Channel diabase. Curve 16 is from a dike and the others are from sheets.

comparable trend, about 3 km to the east. Sites 18 (including baked contacts), 130, and 133 have highly scattered magnetization directions which did not respond to magnetic cleaning. The directions at other sites are initially scattered, but respond well to magnetic cleaning (Fig. 4 Table 2). Sites 16 and 20, from the smaller dike, have inclinations of about −50°, agreeing well with inclinations for the diabase sheets. The 6 good sites from the larger dike have lower inclinations (average about −30°) than the sheets (Table 1). The difference between the two dikes indicates that they did not cool at the same time.

The saturation isothermal remanent magnetizations (i.r.m.(s)) are typical for diabase, the dikes being rather more magnetic than the sheets (Table 3). The steady-field demagnetization curves have a quasi-linear form, and the mean coercivities (about 130 Oe) are also typical of diabase (Table 3). The intensity of the stable component after magnetic cleaning is less than 10% of the initial intensity, the values for the dikes being generally lower than those for the sheets.

Polished sections from each site were examined. Pyrite and pyrrhotite invariably are present in grains that range in size from 100 μ down to a few microns, but by far the most common opaque minerals are large composite grains (50–500 μ) that consist of a cubic mineral intergrown with ilmenite. Small (5μ or less) grains of this cubic mineral occur disseminated throughout the rock, most appearing to replace non-opaque grains, but they are not abundant. The intergrowths in the composite grains occur as irregular, worm-shaped patches or more commonly as lamellae. In a few instances, the lamellae are very thin (3 μ or less) and these sections are all from sites with comparatively high coercivities (sites 23, 55, and 138), but more commonly they have thicknesses exceeding 10 μ. The cubic host mineral of the composite grains has a granulated appearance and ranges in color from light to dark gray; its composition has not been established but it is probably magnetite that has undergone variable alteration to maghemite. The composite grains commonly have corroded margins and internal cracks. In addition to its occurrence as lamellar intergrowths, ilmenite is also present as discrete, pinkish grains marginal to, or enveloped within, the larger composite grains. Leucoxene is a common alteration product.

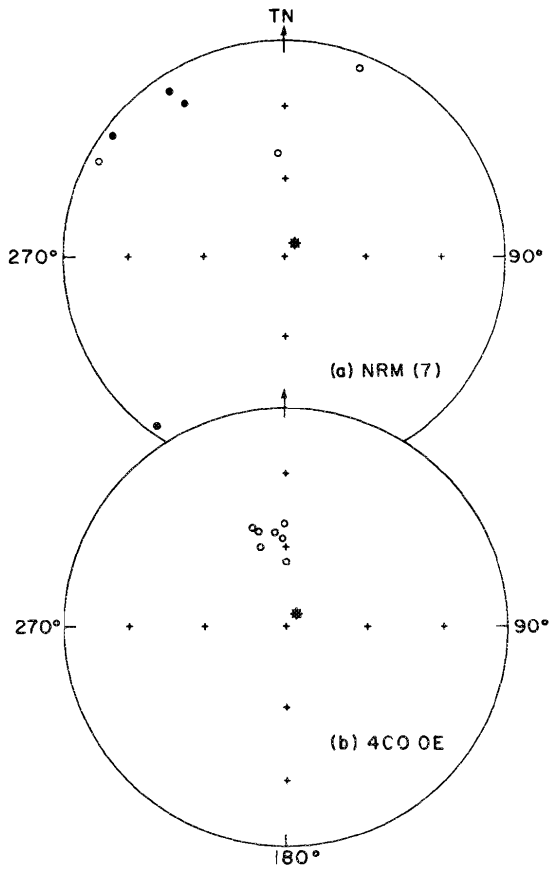


FIG. 6. Magnetic cleaning in diabase (site 7). Seven individual specimen directions from 4 cores are plotted with respect to the horizontal plane. Solid (open) symbols denote positive (negative) inclinations. The present earth's field direction is marked by a star.

The lamellar intergrowth textures in the composite grains are probably the result of high temperature exsolution, but in addition there has been extensive secondary alteration that is likely to have affected the magnetic properties. It is shown below that the remanent magnetization was acquired at the time the diabase cooled and has since remained unchanged; the secondary alteration therefore probably occurred during initial cooling.

Magnetization of Nearby Host Rocks

Fine-grained sediments were sampled at 3 sites. At sites 14c and 31c, the samples were

TABLE 1. Summary of magnetic properties

							m.d.f.	
	n	R	M_N	Treatment	D, I	Pole	a.r.m.	i.r.m. (s)
Western Channel diabase sheets and contact rocks								
7	5	4.97	233	400	352, −56	13N, 111W	118	206
12	3	2.97	725	300	011, −60	—	100	197
12 c	3	2.99	17	400	013, −63	—	242	448
12 (mean)	6	5.96	—	—	012, −61	19N, 127W	—	—
13	3	2.97	657	300	355, −53	10N, 114W	136	237
14	3	2.96	558	300	355, −60	—	079	160
14 c	3	2.99	95	400	348, −60	—	277	230
14 (mean)	6	5.95	—	—	351, −60	18N, 111W	—	—
15	3	2.97	1550	400	003, −56	13N, 120W	087	150
25	3	2.97	970	300	355, −53	10N, 113W	183	328
26A	3	2.87	748	300	341, −63	22N, 104W	142	228
26B	3	2.97	785	300	337, −56	15N, 099W	147	248
27	3	2.96	601	300	005, −51	09N, 122W	102	207
28	3	2.96	588	300	348, −59	17N, 108W	116	213
29	3	2.97	904	300	355, −59	—	091	192
29 c_1	3	3.00	959	500	001, −62	—	422	558
29 c_1	3	2.99	—	1000	356, −62	—	—	—
29 c_1	3	2.99	—	2900	003, −60	—	—	—
29 c_2	5	4.99	129	500	008, −65	—	—	—
29 c_2	5	4.98	—	1000	007, −56	—	—	—
29 c_2	5	4.87	—	2900	009, −60	—	—	—
29 (mean)	11	10.9	—	—	002, −62	20N, 119W	—	—
30	3	2.95	710	300	318, −57	19N, 083W	138	251
31	6	5.98	537	400	353, −52	—	095	194
31 c	5	4.76	72	400	336, −51	—	295	362
31 (mean)	11	10.70	—	—	345, −52	10N, 105W	—	—
32	3	2.99	1070	300	002, −56	13N, 119W	086	171
33	3	2.96	411	300	357, −67	26N, 116W	147	254
34	3	2.99	449	400	357, −36	03S, 115W	122	215
35	3	2.99	530	300	004, −53	10N, 121W	093	192
36	3	2.94	484	300	016, −75	38N, 127W	098	195
38	3	2.95	991	300	001, −52	09N, 119W	112	230
39	3	2.99	803	300	005, −56	14N, 122W	105	205
40	3	2.99	637	300	352, −69	29N, 112W	118	210
42	3	2.75	909	300	344, −49	07N, 104W	144	248
43	3	2.98	762	400	355, −44	03N, 113W	154	274
55	3	2.94	431	300	351, −40	01S, 109W	338	371
58	3	2.93	632	400	359, −48	06N, 117W	313	360
Western Channel dikes and contact rocks								
16	3	2.95	860	300	358, −49	07N, 116W	120	197
19	4	3.91	547	300	005, −26	10S, 123W	115	221
20	3	2.99	1120	400	002, −51	—	135	249
20 c	3	2.97	2	200	014, −67	—	—	3920
20 (mean)	6	5.89	—	—	007, −59	16N, 123W	—	—
21	3	2.99	233	400	356, −34	05S, 114W	097	194
22	2	2.00	381	400	356, −36	04S, 114W	090	157
23	3	2.98	13400	400	003, −36	04S, 121W	172	303
24	3	2.98	247	300	003, −40	01S, 120W	101	210
129	3	2.98	780	300	349, −23	12S, 107W	090	190
137	2	1.93	1790	300	001, −28	09S, 119W	161	203
138	2	2.00	657	400	003, −35	05S, 121W	187	285
Port radium sill								
144	3	2.99	12400	400	138, +59	21N, 084W	150	298
Hornby Bay group, red sandstone								
37	2	1.97	9	2000	151, +61	22N, 095W	—	—
Cameron Bay group, porphyry								
56	3	2.92	83	300	165, +41	01N, 104W	—	—

The letter (c) denotes samples from the country rock near the diabase contact, n is the number of hand samples or cores (2 specimens measured from each), M_N is the intensity of n.r.m. (c.g.s. $cm^{-3} \times 10^{-6}$), the cleaning field is given in oersteds, D, I are the declination and inclination of the mean site direction giving unit weight to each independently oriented core or hand sample. The western hemisphere pole is given. The median destructive field (m.d.f.) of a.r.m. and i.r.m. is given in the last two columns. R is the resultant.

within 0.5 m of contacts and have been visibly baked. At site 29c, samples were obtained at two levels: the first (c_1) is about 0.5 m from the contact and has been baked; the second (c_2) is 8 m from the contact and the rock is not visibly altered. Porphyritic lavas near diabase were sampled at 2 sites, one (site 20c) within 0.5 m of the contact, and another (site 12c) from a thin flow within a dominantly sedimentary section at a distance of 6 m from the contact. The a.f. and thermal demagnetization curves (Figs. 7 and 8) are generally of a stable type, commonly showing an initial increase (by more than a factor of 2 at site 31) which is characteristic of a stable component with negative (reversed) inclination upon which a component of low stability due to the present field is superimposed. The coercivity of contact material is about twice that in diabase (Table 3). Furthermore, contact rocks show very high directional stability: for example, the mean directions at $29c_1$ and $29c_2$ are not significantly changed after demagnetization in 2900 Oe (Table 1), and the intensity after cleaning is reduced by only 30% compared to over 90% for diabase (Table 3). The directions observed in contact specimens are plotted in Fig. 9a. Their mean directions are not significantly different from those in adjacent diabase (Tables 1 and 4).

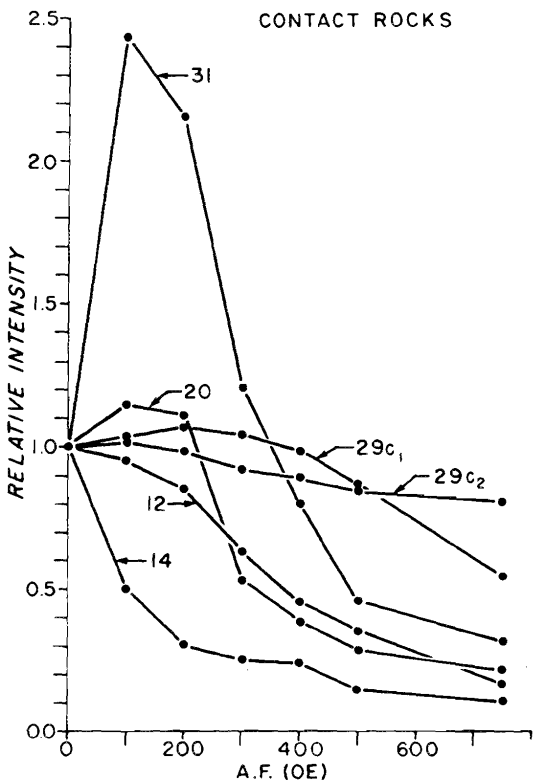


FIG. 7. Alternating field demagnetization curves for rocks near diabase contact. Peak field in oersteds is given.

TABLE 2. Average within-site dispersion

	N	Standard deviation (°) n.r.m.	Cleaned
Sheets	25	48	11
Dike	10	40	9
All diabase	35	46	10
Contacts	6	30	8

N is the number of sites. The dispersion is measured by the circular standard deviation.

The sediment at site 29 has blocking temperatures of 500 and 650 °C and the magnetization is therefore probably due to both magnetite and hematite. This is confirmed by study of polished sections, which show fresh magnetite (20–200 μ subhedral and skeletal grains) and specular hematite (2–100 μ grains containing flake-like intergrowths). The other contact rocks for which thermal demagnetization has been carried out have blocking temperatures predominantly in the range 500–550 °C (characteristic of magnetite); polished sections show very small (3μ) opaque minerals that have not been identified. Polished sections of baked lava at site 20 with coercivities in excess of 3000 Oe show specular hematite (20μ and less), leucoxene, and minor pyrite.

TABLE 3. Average coercivities and magnitudes of stable components

	N	i.r.m. (s)	m.d.f. a.r.m.	i.r.m. (s)	M/M_N
Diabase sheets	25	0.675	135	230	0.092
Diabase dikes	10	0.892	127	221	0.073
All diabase	35	0.737	132	225	0.087
Contact rocks	4	0.018	309	400	0.713

N is the number of samples averaged (one per site). The i.r.m. (s) is the saturation isothermal remanence. M/M_N is the ratio of the magnetization intensity after cleaning to that of n.r.m. Coercivities are estimated by the median destructive field (m.d.f.). In the last row results from 20c are omitted because the coercivity is so high (Table 1) that its inclusion would unfairly dominate the average.

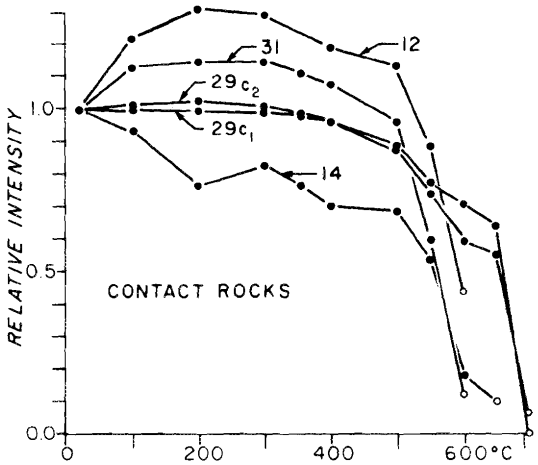


FIG. 8. Thermal demagnetization of rocks near diabase contact. Open circles denote the temperature at which the directions become random. Results from baked porphyry near the dike at site 20 are not plotted because the relative intensity increases to over 2 in the range 100 to 400 °C.

To study heating effects at greater distances from contacts, samples of green and red tuffaceous sediments were obtained at distances between 15 and 60 m beneath diabase (site 26). Some directions are widely scattered, but others are approximately in the same direction as the diabase (Fig. 9b). Evidently within this distance range, partial, and in some instances complete, remagnetization has occurred. At sites more than 1 km from contacts, the directions bear no relationship to those in the diabase (Fig. 9c), nor do the rocks show signs of being reheated by the diabase.

Discussion of Data from Diabase and Nearby Host Rocks

The results for the diabase and nearby host rocks show that reheating sufficient to remagnetize rocks with blocking temperatures as high as 650 °C (Fig. 8) has occurred at distances of up to 6 to 8 m from contacts. Comparable remagnetization effects occur sporadically up to a distance of 60 m, but have not been found beyond 1 km from the diabase. Thus the remanent magnetization directions of the diabase (Fig. 4a) were acquired at the time of intrusion and cooling, and almost certainly represent the geomagnetic field at that time. The data (rows 7 and 8 of Table 4) summarize results based on 129 oriented samples (usually 2 specimens from each). Sites are given unit weight (N = 35), but because some sites may have become magnetized at essentially the same time (the dike rocks in particular), each may not be an independent estimate of the field direction, so that errors calculated in this way are too small. The values in brackets were calculated assuming that half the sites are statistically redundant (N = 18) and these values are considered to be more realistic estimates of errors in mean direction and pole position.

Port Radium Sheet

Samples were obtained from a diabase sheet that intrudes sediments of the (undifferentiated) Cameron Bay – Echo Bay Groups. The direction of magnetization is stable, being little changed after heating to 550 °C. The square-

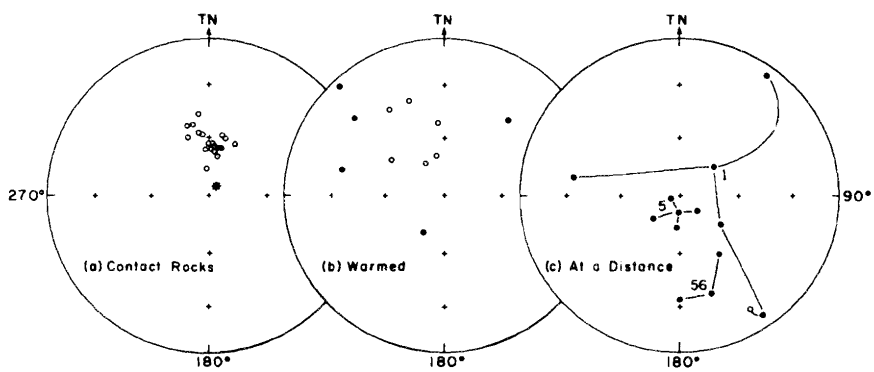


FIG. 9. Heating effect of diabase. In (a) individual specimen directions (cleaned) in sediments and lavas less than 8 m from diabase contacts are plotted. In (b) directions (cleaned) in cherty siltstones at distances of 15 to 60 m from contacts are shown. In (c) directions of volcanics (site 1 cleaned at 200 Oe, and site 56 cleaned at 300 Oe) and argillite (site 5 cleaned at 1500 Oe) are plotted. The primitive is the horizontal and the present earth's field is marked by a star.

TABLE 4. Average directions of magnetization and poles in Western Channel diabase and contact rocks

	N	Coordinates	R	k	α_{95}
1. Directions in contact rocks	5	357, −62	4.94	62	10
2. " " diabase near (1)	5	359, −57	4.98	181	6
3. " " dikes	10	356, −36	9.86	66	6
4. Poles for dikes	10	03S, 118W	9.90	69	5
5. Directions in sheets	25	355, −56	24.60	60	4
6. Poles for sheets	25	14N, 113W	24.40	40	5
7. Directions in all sites	35	**356, −50**	34.03	35	**4(6)**
8. Poles for all sites	35	**09N, 115W**	33.91	32	**4(6)**

N is the number of sites. In the second column the coordinates of the average directions (D, I) and poles (lat., long.) are given. R is the resultant, k is Fisher's precision, and α_{95} the radius of the circle of confidence in degrees. In lines 7 and 8 the averages of all site results (including contacts) are given; the values in bold face are regarded as the best summary of the data (see text).

shouldered demagnetization curve (Fig. 10) with blocking temperatures between 500 and 560 °C indicates that the remanence is due to magnetite. Study of polished sections showed that the opaque minerals consist of composite grains (up to 500 μ) of magnetite (characterized by somewhat granulated surfaces) and ilmenite. The ilmenite is present as lamellar intergrowths, as surface coatings, and as irregular grains embedded within the magnetite groundmass. The directions (Fig. 11) are toward the southeast, with positive inclination (*i.e.* approximately reversed from that in the Western Channel diabase).

Hornby Bay Group Sediments

The sandstones of this group are, for the most part, white, light gray, or pink. Fine-grained siliceous red sandstones are abundant as slabs on beaches, but were found *in situ* only at two places (sites 37 and 57). Specimens from site 57 were weakly magnetized and the directions random. Specimens from site 37 gave excellent square-shouldered demagnetization curves (Fig. 10) with blocking temperatures in the range 650–700 °C, indicating that the magnetization is due to hematite. This was confirmed by a study of polished sections that show very fine-grained red material (presumably hematite) and abundant semi-rounded particles of specular hematite (10–30 μ) that are probably detrital. The directions are internally consistent and indistinguishable from those observed in the Port Radium diabase (Fig. 11).

Cameron Bay – Echo Bay Groups

In addition to the previously described study of baked sediments near diabase contacts,

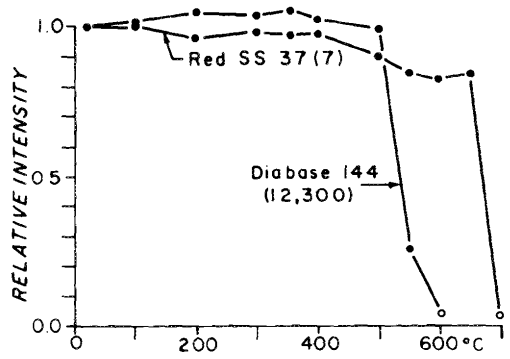


FIG. 10. Thermal demagnetization of Port Radium diabase (site 144) and red sandstone of the Hornby Bay Group (site 37). Initial intensities (c.g.s. $cm^{-3} \times 10^{-6}$) are given in brackets. Open circles denote the temperature at which the directions become random.

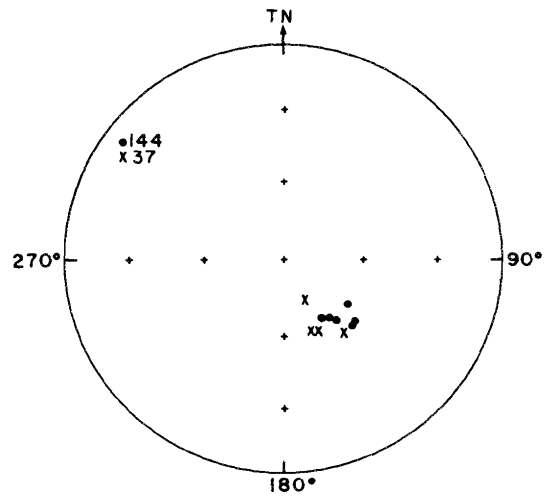


FIG. 11. Directions of magnetization in Port Radium diabase (site 144 cleaned at 400 Oe) and in red sandstone of the Hornby Bay Group (site 37 cleaned at 600 °C).

samples were obtained from 7 sites (1, 5, 41, 56, 134, 140, 143) more distant from diabase and representing a variety of rock types. The results are disappointing. The directions at sedimentary sites 5, 41, 134, 140, and 143 are either widely scattered, or are close to the present field direction. An example (site 5) is shown in Fig. 9c.

Volcanics are abundant in the Cameron Bay Group, but are commonly massive and their original attitudes could be determined only at sites 1 and 56. An age of 1770 ± 30 m.y. (Rb–Sr isochron, Robinson and Morton 1972) has recently been obtained for these volcanics. At site 1 the directions are random and do not respond to cleaning. Site 56, in porphyritic lava that shows sub-horizontal flow banding, has southeast directions, similar to those observed in both the Hornby Bay sandstone and the Port Radium diabase.

Samples from diorite (Fig. 3) intrusive into the Cameron Bay – Echo Bay Groups (sites 131 and 132) are unstable.

Discussion

This study has provided a reliable pole for the Western Channel diabase. Results from other rocks are generally disappointing, but southeast directions were found in Port Radium diabase, Hornby Bay sandstone, and lava of the Cameron Bay Group. Although the latter results are represented by only 3 sites, and are consequently of a preliminary nature, they do show agreement between different rock types and stable magnetizations. The rocks are all older than the Western Channel diabase, and the fact that their directions are very different (they are approximately reversed from one another (compare Figs. 4b and 11)) suggests that their magnetization predates that of the diabase.

The poles derived from this work, and other observations from rocks of lower Proterozoic and Archean age from the Canadian Shield, are plotted in Fig. 12 together with their approximate ages. Poles in the age range 1250–1550 m.y. fall in the central Pacific Ocean. The results reported herein and those recently obtained from the Et-Then Group and the Dubawnt Group indicate that the pole then (proceeding backwards in time) moved eastwards in a broad sweep to the vicinity of what is nowadays northern South America. Results from oldest Proterozoic (Aphebian) and Archean rocks are scanty, but they can be connected by a return loop into the southeast Pacific as shown in Fig. 12. These results, together with those already published for younger Proterozoic rocks (the polar path is given in Fig. 12) indicate that during the Proterozoic, the path of apparent polar wandering relative to the Canadian Shield made several loops, with peak-to-peak amplitudes of approximately 90° and periods of several hundred million years. The significance of this for the correlation of Precambrian rocks and for the tectonic history of the Canadian Shield will be considered in a subsequent paper (Donaldson *et al.* 1972, in press).

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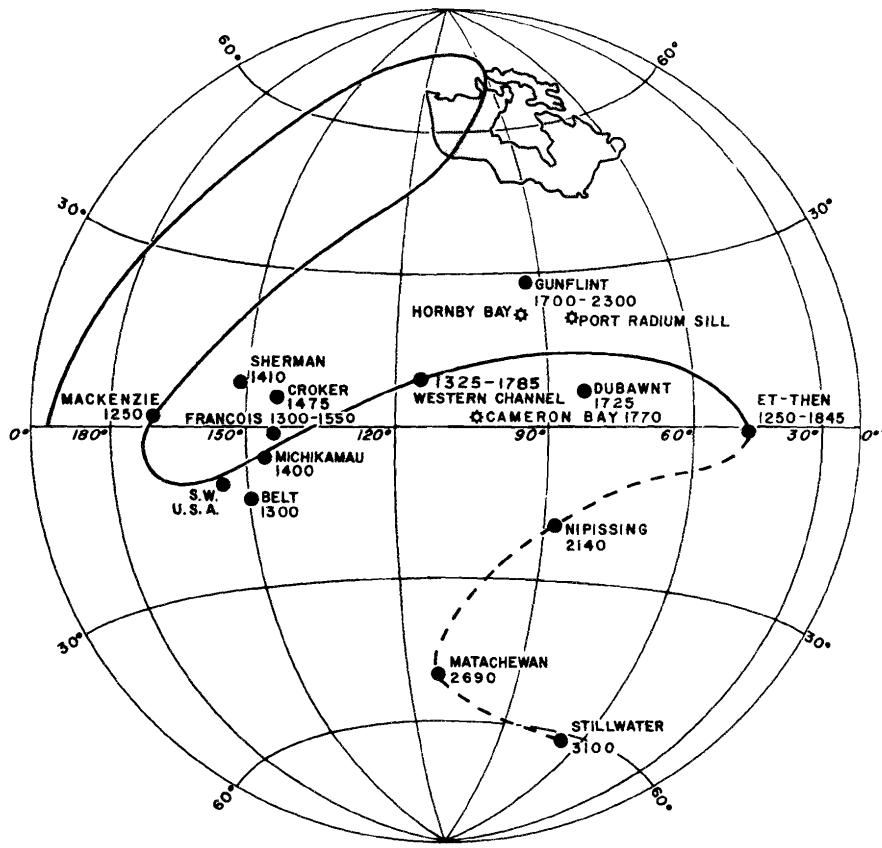


FIG. 12. Pole positions for Lower Proterozoic and Archean rocks of the Canadian Shield (shaded). Solid dots are results from many sites, open symbols being the results from single sites reported here. Stillwater (Bergh 1970); Matachewan dikes (average of results of Fahrig *et al.* 1965, and Strangway 1964); Nipissing diabase (Symons 1970); Et-Then Group (Irving *et al.* 1972); Dubawnt Group (Park *et al.* 1972, submitted for publication); Gunflint Formation (Symons 1966); Croker Island Complex (Palmer 1969); Michikamau anorthosite (Murthy *et al.* 1968); Beltian rocks (average of results of Collinson and Runcorn 1960, Norris and Black 1961); sediments of southwestern U.S.A. (average of results of Collinson and Runcorn 1960); St. Francois igneous rocks (average of results of Hays and Scharon 1966, Chi-Hsu *et al.* 1966); Mackenzie igneous rocks (average of results of Robertson 1969, Fahrig and Jones 1969, Larochelle 1966, Irving *et al.* 1972); Sherman Granite (Eggler and Larson 1968) this pole (08S, 151W) has been incorrectly plotted but without affecting the trend of the curve. The Upper Proterozoic pole path, as developed by DuBois (1962), Black (1963), and Robertson and Fahrig (1971), is also given.

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